Week 0:
Course Introduction

Ye WANG

Course Format

- **Course ID: FTE4312**
- **Course Name: Blockchain and Decentralized Applications**
- **Instructor: Ye WANG (kanyewang@cuhk.edu.cn)**
- **Lecture Hours: 3 Hours per week @ *TxB 201***
  - **15:00 – 18:00 Wednesday**
- **Office Hour: Appointment via email.**

Teaching Assistants:



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What can we learn from this course?

Your Expectation?



Course Description

- The course will cover the technical and business aspects of blockchain technologies, distributed system, bitcoin, smart contract and Ethereum virtual machine, cryptocurrencies, non-fungible tokens, and decentralized applications, including decentralized finance (decentralized exchange, lending, liquid staking, yield farming, derivatives, oracles, etc.), Web3 security, etc. Students will learn how these systems work, how to develop smart contract based software application, how to analyze a decentralized application.

Prerequisite

- Basic Programming Languages, such as Python, JavaScript, C/C++, etc.
- Basic Data Structure, Computer Network, Distributed Computing

Learning Outcome

The objective of the course is to impart a working understanding of blockchain and decentralized applications to students. After complete this course, the students are required

- To develop an understanding of blockchain and consensus models
- To develop an understanding of decentralized ledger implementation
- To perform solidity smart contract design and implementation
- To develop an understanding of cryptocurrencies and NFTs
- To perform analysis on decentralized markets
- To develop an understanding of privacy and security in blockchain
- To develop an understanding of performance optimization in blockchain

Textbook

- Course slides
- Cryptocurrency whitepapers
- Online Docs and Tutorials
  - Solidity by Example
    - https://solidity-by-example.org/
  - Ethernaut
    - https://ethernaut.openzeppelin.com/
  - CryptoZoombies
    - https://cryptozombies.io/

Assessment Scheme

Component/ method	% weight
Assignments	**20%**
Midterm Exam	**15%**
Final Exam	**30%**
Final Project	**35%**

2 Assignments, 10% each

Close book Wrtten Test (Topic 1 to Topic 5)

Close book Wrtten Test (Topic 1 to Topic 11)

Group Project: 1~2 members in a team
Create a blockchain project with Solidty Smart Contact:
Story: 15%, Implementation: 10%, Presentation 10%

Course Schedule (Subject to Changes)

Week	Topics	Assignments
1	0. Course Intro + 1. Distributed System and Consensus	
2	2. BitCoin	
3	3. Ethereum and Smart Contract	
4	4. Cryptocurrency and NFT	
5	5. Solidity Development and Review	**Assignment 1**
6	Midterm	
7	6. Decentralized Exchanges and Lending	
8	7. Blockchain Investment and Arbitrage	
9	8. Markets, Trading & Data	
10	9. Blockchain Security	**Assignment 2**
11	10. Bridge, Layer 2, and Other Consensus	
12	11. Guest Lecture (Prof. Ruizhe Jia from Stanford)	
13	Project Presentations	
14	Project Presentations + Review	
15	Final Exam	

January - 2026	29	30	31*	1	2	3	4班
	5	6	LEC	8	9	10	11
	12	13	LEC	15	16	17	18
	19	20	LEC	22	23	24	25
	26	27	LEC	29	30	31	1
February - 2026	2	3	LEC	5	6	7	8
	9	10	11	12	13	14	15
	16	17春节	18	19	20	21	22
	23	24	25	26	27	28	1
March - 2026	2	3	MT	5	6	7	8
	9	10	LEC	12	13	14	15
	16	17	LEC	19	20	21	22
	23	24	LEC	26	27	28	29
April - 2026	30	31	LEC	2	3	4	5清明节
	6	7	LEC	9	10	11	12
	13	14	LEC	16	17	18	19
	20	21	PRE	23	24	25*	26
May - 2026	27	28	LEC	30	1劳动节	2	3
	4	5	Final	7	8	9	10
	11	12	13	14	15	16	17

Week 1 Distributed Systems & Consensus

- Why blockchain matters
- Basics of distributed systems
- Consensus problems
- Overview of classical consensus mechanisms

Week 2 Bitcoin

- Bitcoin architecture and transaction model
- Proof-of-Work and mining incentives
- Security assumptions and limitations

Week 3 Ethereum and Smart Contracts

- Ethereum’s account-based model
- Ethereum Virtual Machine (EVM)
- Smart contracts
- Proof-of-Stake

Week 4 Cryptocurrency and NFT

- Coin and token
- Token standards (ERC-20, ERC-721, ERC-1155)
- Stablecoin: mechanism and security issues
- NFTs: creation, ownership, and metadata

Week 5 Solidity Development

- Introduction to Solidity language
- Smart contract structure and syntax
- State variables, functions, and modifiers
- Common development patterns
- Basic security pitfalls in smart contract coding

Week 7 Decentralized Exchanges and Lending

- Automated Market Makers (AMMs)
- Liquidity provision and pricing mechanisms
- Decentralized lending protocols
- Collateral, liquidation, and interest rates

Week 8 Blockchain Investment and Arbitrage

- Synthetic assets: design, collateralization, and oracle dependence
- Perpetual futures, options, and leverage mechanisms
- Miner / Maximal Extractable Value (MEV): Arbitrage, Sandwich, and Liquidation

Week 9 Markets, Trading & Data

- Key players in crypto finance
- Market games in blockchain systems

Week 10 Blockchain Security

- Economic vs technical security
- Smart contract vulnerabilities
- Common attacks (reentrancy, frontrunning, MEV)
- Auditing and formal verification

Week 11 Bridge, Layer 2, and Other Consensus

- Scalability challenges of blockchains and Layer 2 solutions
- Cross-chain bridges and interoperability
- Future directions of blockchain consensus

Blackboard System

- Slides will be posted to Blackboard system
- Let me know if you can not access the resources ASAP!

Policy on Plagiarism

- ZERO tolerate to plagiarism!
- You need to complete your project, exams all by yourself (except the group project)!
- You will directly receive an F if you cheat!